TIC 237505178

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period        5.186476 d
  Epoch (t0)    1470.6479 BTJD
  Duration      0.1700 d
  Depth         3.501 %  (35,014 ppm)
  SNR           77.6

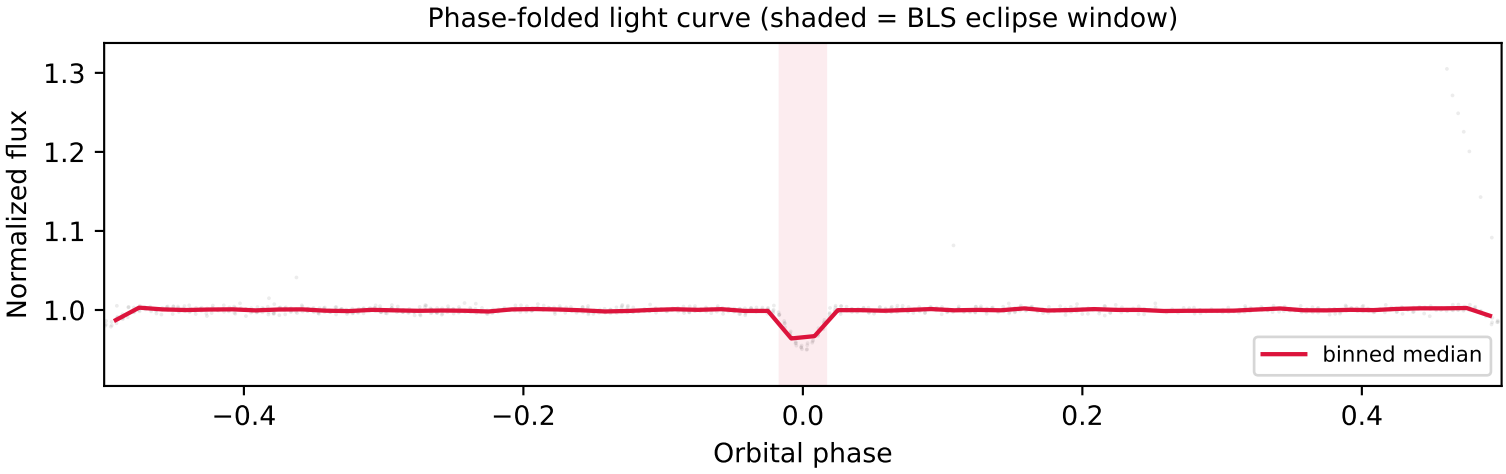
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      4  (gate >= 3)
  Robust depth ratio     0.94  (gate >= 0.50)
  Robust depth sigma     47.1  (gate >= 7.0)
  Odd/even z-score       0.30  (consistent)
  Secondary eclipse      detected, phase -0.49, 5.9 sigma
```

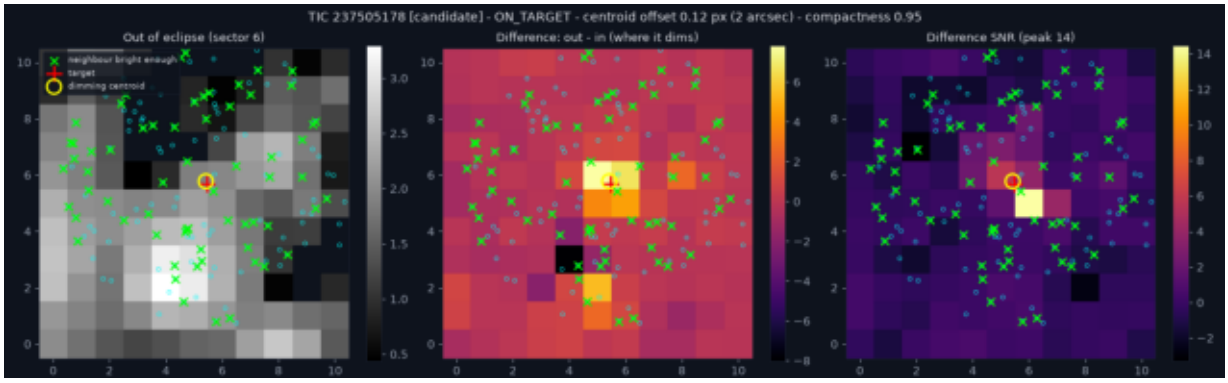
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CATALOG CROSS-MATCH (novelty)

  VSX                                  no match
  ASAS-SN                              no match
  Gaia eclipsing-binary table          no match
  Gaia variable classification         no match
  TESS EB catalog (Prsa+2022)          no match
  ExoFOP TOI list                      no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 6    light-curve dip in this sector: 48.2 sigma    t0 refined by -0.0050 phase
  Difference-image peak SNR   14.47   (gate >= 5.0)      Compactness   0.95   (gate >= 0.50)
  Dimming centroid offset     0.12 px (2 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 237505174
    0.43 px away, dTmag +0.01 - farther from the dimming than the target
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No caveats - clean detection on every check.